Lesson 2: Debugging with AI

Module 1 | 60 minutes | Describe · Share · Test · Understand | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. bug ___ A. The program runs but gives the wrong answer.
b. debugging ___ B. Python's note about what went wrong and on which line.
c. error message ___ C. Finding and fixing bugs.
d. logic bug ___ D. A mistake in a program that makes it go wrong.

2. Prompt Lab: the perfect bug report

Write a bug report prompt for Meera's score bug. Include what you expected, what happened, the exact error message with the line number, what you already tried, and ask for a hint before the full answer. Write your prompt here, then try it in Kitabash AI helper or ChatGPT or Gemini or Copilot.

3. Match each error to what it usually means

Draw a line from each word to its match.

a. SyntaxError ..... A. No error, but the answer is wrong
b. NameError ..... B. The spaces at the start of a line are wrong
c. TypeError ..... C. Mixing types that cannot go together, like text + number
d. IndentationError ..... D. A name is not defined — often a typo
e. Logic bug ..... E. Python cannot read a line — a missing bracket or quote

4. Helpful bug report or unhelpful?

Write each item in the right group: Helpful / Unhelpful.

a. “I expected 77.75 but got 103.67.” → _____
b. “It’s broken, fix it.” → _____
c. “NameError: name ‘scoer’ is not defined, line 4.” → _____
d. “I already tried changing the loop to range(4).” → _____
e. Pasting 300 lines of code with no question → _____
f. “Just send me the right program.” → _____

5. Plan your program

Before you type it: Debug it! This program should print the average of four test marks: `Average: 77.75`. It has three bugs: one TypeError and two logic bugs. Fix them. Write the steps in plain English (pseudocode), then the Python you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can write a bug report with expected, actual, error and what I tried.
☐ I can test an AI’s fix instead of trusting it.
☐ I can tell an error from a logic bug.
☐ I can explain why a bug happened in one sentence.